



Management Science

Publication details, including instructions for authors and subscription information:
<http://pubsonline.informs.org>

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Wei Wang; , Chris Yung

To cite this article:

Wei Wang; , Chris Yung (2025) Employment Protection and Venture Capital Investment: The Impact of Wrongful Discharge Laws. Management Science 71(2):1523-1545. <https://doi.org/10.1287/mnsc.2023.01936>

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Employment Protection and Venture Capital Investment: The Impact of Wrongful Discharge Laws

Wei Wang,^a Chris Yung^{b,*}

^aMonte Ahuja College of Business, Cleveland State University, Cleveland, Ohio 44114; ^bMcIntire School of Commerce, University of Virginia, Charlottesville, Virginia 22904

*Corresponding author

Contact: w.wang24@csuohio.edu (WW); cay2m@virginia.edu,  <https://orcid.org/0000-0002-9690-3735> (CY)

Received: June 26, 2023

Accepted: September 18, 2023

Published Online in Articles in Advance:
May 22, 2024

<https://doi.org/10.1287/mnsc.2023.01936>

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Abstract. Wrongful discharge laws (WDLs) provide limits to the employment-at-will doctrine, and thus impair operating flexibility, increasing expected financial distress costs by making it costly to fire employees. This impairment is detrimental to start-ups, leading to a decline in venture capital (VC) investment. Using a difference-in-differences framework enabled by the staggered adoption of WDLs across the U.S. states, we show VC investment declines after a state adopts the good faith exception (the strongest form of WDL). This decline is most pronounced in sectors with high labor dependency.

History: Accepted by Victoria Ivashina, finance.

Supplemental Material: The data files are available at <https://doi.org/10.1287/mnsc.2023.01936>.

Keywords: wrongful discharge laws • venture capital • difference-in-differences

1. Introduction

Firms value operating flexibility—the ability to scale operations up or down when conditions warrant—because it facilitates the exercise of real options. For example, consider a firm engaging in research and development (R&D) for an innovative new product. If R&D outcomes are encouraging, the firm can hire additional workers to ramp up production. Otherwise the firm scales down by firing workers.

Operating flexibility is particularly valuable for start-ups and other young firms whose value is mostly composed of growth options. Yet operating flexibility can be impaired by labor law. Legal obstacles to firing impose an ex ante cost that employers must consider when hiring new workers. Legal obstacles to labor flexibility reduce the value of firm growth options.

In this paper, we examine the impact of wrongful discharge laws (WDLs) on venture capital (VC) investment in the United States. These laws restrict firms' ability to fire employees by widening the scope for lawsuits if the involved employees view the discharge as unfair. Between 1970 and 1998, different forms of WDLs were adopted by most U.S. states in a staggered manner. In a nonventure setting, Serfling (2016) argues that WDLs limit firms' operating flexibility. Serfling's (2016) concerns should be heightened in a venture capital setting.

We hypothesize that WDLs dampen VC activity. Fewer VC-backed firms are created when the legal environment impedes operating flexibility.¹ Indeed, VCs

often negotiate the right to fire employees as a precondition of financing.

The WDL adoption process in the United States offers two attractive features for research design. First, their staggered adoption enables us to employ the difference-in-differences (DiD) approach to identify the effect of WDLs on VC investment, in a similar spirit as Acharya et al. (2014) and Serfling (2016), who examine, respectively, corporate innovation and capital structure. Second, WDLs are politically motivated by labor relations, for example, the decline of unions and the balance of rights between firms and workers (Walsh and Schwarz 1996). This makes the regulation shift plausibly exogenous in a VC setting because the law changes did not have VC markets as a focal point. Thus, the effect of WDLs on VC investment potentially represents a causal link.

The literature categorizes WDLs into three different types; these are described in more detail in Section 2.1. Listed in order of legal strength, that is, from most binding on the employer to least binding, these types are (i) the "good faith" (GF) exception, (ii) the "implied contract" (IC) exception, and (iii) the "public policy" (PP) exception (Dertouzos and Karoly 1992, Kugler and Saint-Paul 2004, Autor et al. 2007).

As expected, we find that WDLs decrease VC investment at the state level. In particular, the good faith exception exhibits a strong negative effect on VC activity. During the 1980–2003 sample period, states that adopted the good faith exception observed a relative

decline in both the number and dollar amount of VC investment deals compared with nonadopting states. For a state-year with average VC deals, the percentage decline is approximately 22% in the number of deals and 35% in the dollar amount. This effect is robust regardless of the selection of sample and sample period. At the state-sector level, a strong negative effect is also observed. Sector-by-sector examinations show that the negative effect concentrates in several sectors such as biotech, information technology, and communications. These sectors feature low asset tangibility, high R&D intensity, and high market-to-book ratios, all associated with high labor dependency. We also obtain some results that indicate that IC exception seems to depress VC investment, although the effects are generally small and often not statistically significant. The weakest WDL exception, that is, public policy, has no effect on VC investment.

We attempt to establish potentially a causal link between the WDLs and VC investment as follows. First, prior studies (e.g., Acharya et al. 2014, Serfling 2016, Bai et al. 2020) argue that WDLs act as exogenous events to specific corporate policies. A similar argument holds here that WDLs are motivated by equality considerations (labor protection) rather than impact on entrepreneurship. Second, our examination of the dynamic effects of WDLs indicates VC investment declines *after*, but *not before*, the adoption of the good faith exception. Third, survival analysis based on the Cox proportional hazard model also shows that the causality does not go from VC investment to WDL adoption. Fourth, we generate counterfactual samples by randomly assigning WDL exceptions to states in random years and find these hypothetical WDL events do not have a bearing on subsequent VC investment. Such a falsification test provides a counterfactual to show that the negative WDL–VC relationship is not random.

Our finding that WDLs depress VC investment presents an interesting complement to Acharya et al. (2014), which documents an increase in firm creation following the enactment of WDLs, especially the good faith exception. To reconcile the results, we show that although overall more firms are created subsequent to the good faith enactment, the percentage of new firms that eventually receive VC funding drops substantially. This indicates that WDLs encourage the creation of certain types of firms but discourage the creation of firms that are targets of VC investments. Consistent with the conjecture, we find that the number of new firms increases only in the smallest size bucket, with less than 10 employees, which encompasses both self-employed workers and small shops such as caterers and plumbers, among others. In addition, firm creation is heightened only in non-high-tech industries.

1.1. Relationship to the Literature

Our main hypothesis complements another important hypothesis found in the literature, that WDLs promote entrepreneurship. Acharya et al. (2014) argue that these laws incentivize innovation by employees who become less concerned about the possibility of holdup by the firm, and more generic innovations lead to the creation of new firms. In line with this, they demonstrate that the enactment of the good faith exception leads to an increase in innovation activities, measured by patent filings, as well as the establishment of new companies, based on U.S. Census data. Their results appear in conflict with our finding that VC activity declines.

To understand this difference, consider that WDLs are likely to be viewed by workers and owners in an opposite way. From the perspective of a current employee, the WDL passage is reassuring and increases their safety. WDLs are viewed more unfavorably by owners, because reduced operational flexibility diminishes the firm's value. VCs clearly fall into the category of owners, thus leading to a reduction in the flow of VC funding into the market.

Entrepreneurs (or entrepreneurial teams) embody a hybrid role. They are literal employees of the firm, and are therefore concerned about the possibility of being fired by other owners (as exemplified by the case of Cisco's founders being dismissed by their VCs just before the dot-com boom). Yet entrepreneurs are also partial owners who stand to benefit from the operational flexibility that WDLs curtail. Consequently, the impact on entrepreneurs remains ambiguous, contingent on whether for incentive purposes they function more like employees or owners.

The net effect is positive only when the positive effect on employee-driven innovation outweighs the decline in operational flexibility. In our analysis, we replicate the approach of Acharya et al. (2014) and confirm that following the passage of WDLs, more firms are created. Yet we also find that (1) fewer VC-backed firms are created, both as a raw number and as a fraction of overall firms, and (2) firm creation increases only in the smallest size buckets and in non-high-tech industries. This appears to indicate that WDLs do not increase the demand from the entrepreneurs for VC funding. Therefore, the decline of VC investment ensues as VC funds shun elevated operational risk subsequent to the WDL adoption.

Our study is also closely related to that by Bozkaya and Kerr (2014), which examines the determinants of VC activity across European Union (EU) nations. Among other variables, their analysis includes an Organisation for Economic Co-operation and Development (OECD) measure called the "employment protection index," which proxies for the difficulty in firing employees. They find a negative relationship between this index and VC activity. In other words, countries with

stringent labor laws (like Italy) exhibit less VC activity compared with countries with more flexibility, such as the United Kingdom. However, the relationship is to a large degree restrained to the cross-sectional level by the data because the OECD measure has virtually no intertemporal variation, as the sample lacks any labor-flexibility policy shocks.

Our analysis instead benefits from U.S. state-level data that exhibit substantial variation both cross-sectionally and along the time series dimension. The staggered introduction of WDLs across the U.S. states permits a two-way fixed effect (TWFE) difference-in-differences analysis, more cleanly indicating a causal relationship. Moreover, the diverse legal systems² across European countries in the EU sample introduce confounding effects on VC investment. In contrast, the U.S. states adhere to a common set of federal laws.

Our work contributes to the understanding of what shapes the variability of VC investment. Gompers and Lerner (1998) show that capital gains taxes affect the supply of VC funds. Black and Gilson (1998) emphasize the importance of well-developed stock markets and initial public offerings (IPOs) for venture capital financing. Jeng and Wells (2000) confirm that IPOs is one of the most prominent drivers of VC investment in a cross-country study. They also show that average tenure on the workers—considered a proxy for labor market rigidities—is negatively associated with early stage VC activity. Gompers et al. (2008) show that more experienced venture capitalists respond to favorable public market signals earlier and view this as evidence that investment opportunities drive venture capital investment. Cumming et al. (2010) highlight the role of legal origins and accounting standards in explaining cross-country variations in VC development. Cumming and Li (2013) examine a host of institutions and political factors, including minimum wage, unemployment insurance, labor union density, government grants, academic R&D, and bankruptcy laws.

Our findings also add to the literature on interactions between labor economics and finance, particularly how labor policies impact corporate decisions and entrepreneurship. Strong labor rights prompt firms to choose high leverage to strengthen their bargaining position (Matsa 2010), cut dividend payout (Haw et al. 2018), and reduce shareholder gains in acquisitions (John et al. 2015), among others. Higher unemployment benefits increase the use of firm debt (Agrawal and Matsa 2013), decrease in cash holdings (Devos and Rahman 2018), and increase earnings management (Dou et al. 2016). In particular, Hombert et al. (2020) and van Doornik et al. (2022) show that unemployment insurance spurs entrepreneurial activity. Also, states that enforce noncompete covenants experience less vibrant creation of new firms and a lower return on venture capital investment (Samila and Sorenson 2011, Starr et al. 2018, Jeffers

2024). The adoption of antitroll laws enhances venture capital funding and patent-based lending to high-tech start-ups (Appel et al. 2019). WDLs discourage the use of debt (Serfling 2016) and capital investment (Bai et al. 2020), but encourage innovation as proxied by patents and new production methods (Acharya et al. 2014, Bena et al. 2022).

Turning to our policy implications, labor market policies may involve a trade-off between efficiency and equality. Employment protection laws such as WDLs effectively tax employers in the form of increased labor costs, which is especially detrimental to start-ups burdened by the liabilities of newness, smallness, and novelty (e.g., Stinchcombe 1965, Freeman et al. 1983, Amason et al. 2006, Hyytinen et al. 2015). Externalizing labor protection via unemployment insurance provides employee protection while not discouraging venture capital activity. This finding complements other research that finds that the effect of unionization leads to reduced R&D and a decline in patent quality (Bradley et al. 2017), showing another channel through which impaired operating flexibility potentially affects innovation.

The rest of this paper is organized as follows. Section 2 provides the background of WDLs. Section 3 describes data and methodology. Section 4 presents the main results at the state level. Section 5 extends the investigation to the state and sector level. Section 6 concludes.

2. Wrongful Discharge Laws

2.1. Adoption of WDLs

The “at-will employment” doctrine used to define the labor market in the United States grants employers the unencumbered ability to dismiss workers in response to changes in their operating environments. The arbitrary nature of at-will employment had long been criticized (Blades 1967). In the 1970s, state judiciaries begin enacting “exceptions” that provided employees a legal channel to challenge dismissal decisions that occur without a “just cause.” This process accelerated in 1980s, a period that coincided with the decline of unions, and a recessionary economy that caused high unemployment rates. These exceptions, often known as wrongful discharge laws collectively, largely fall into three baskets: good faith, implied contract, and public policy.

The good faith exception is rooted in the tenet that neither the employer nor the employee should be allowed to take unfair actions that terminate the employment. Specifically, courts require employers to establish that dismissal decisions are in “good faith.” The interpretation of good faith varies from case to case. Generally, the employer is expected to have followed a reasonable rule for the dismissal that had been applied evenly to all employees, for example, conducted a timely and thorough investigation of the employee’s conduct and provided the employee advanced notice.

The implied contract exception states that the employment cannot be terminated by the employer without a good cause in the presence of implicit, enforceable promises of job security. In other words, even if no express representation exists in a written contract, the employee can cite employee handbooks, oral statements made by supervisors or interviewers, the history of the employer's practices with respect to retaining employees, and the like to challenge the dismissal decisions of the employer.

The public policy exception offers an employee the cause of action against the employer's dismissal decision when the dismissal would contravene some well-established public policy. For example, the employee should not be terminated for reporting the employer's wrongdoing or refusing to commit an illegal act. The principle behind this exception is that the public good outweighs private interests of an organization, and an employee should not be penalized for acting in a way to promote the public good.

The implied contract exception is the most widely adopted, followed by the public policy exception. As of the turn of the century, 41 states had adopted the implied contract exception, 43 states had enacted the public policy exception, but only 12 states had established the good faith exception.³ Figure 1 shows the staggered adoption of these WDLs. As is evident in the figure, 1980s witnessed a rapid increase in the adoption of the implied contract and public policy exceptions,

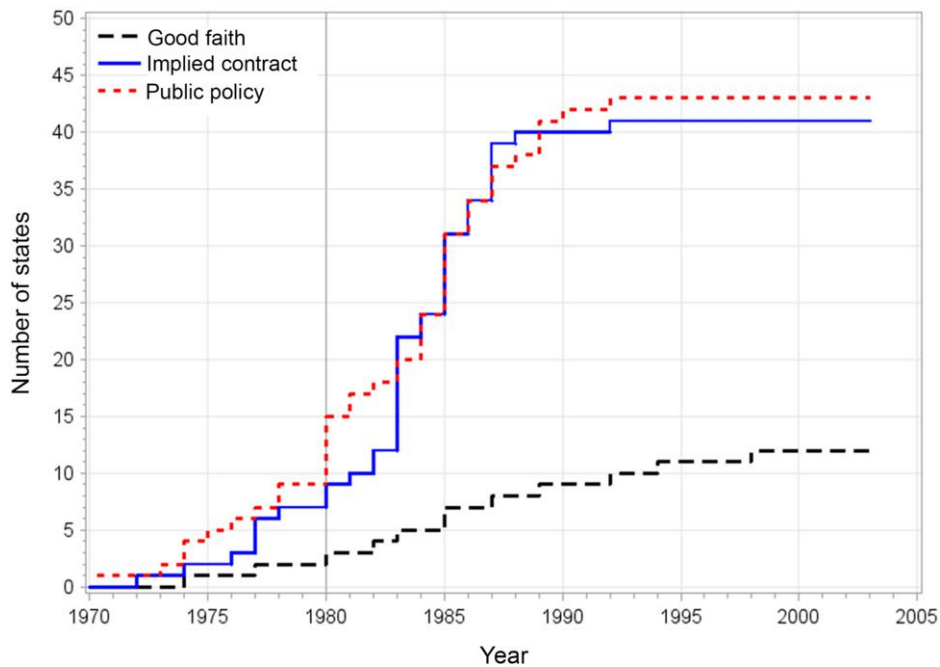
and the adoption of the good faith exception was slower and extended well into 1990s.

2.2. Impacts of WDLs

WDLs curtail the at-will employment doctrine and restrict employers' flexibility in firing workers. As a result, they effectively impose additional labor costs on employers. Direct costs include, and are not limited to, damage awards to discharged employees who sue. According to *Newsweek*, in trials of wrongful discharge cases, verdicts run heavily in favor of employees, as sympathetic jurors often perceive big business as the bully. "Employees won 72 percent of the libel and other job-termination trials" in the seven-year period of 1980–1986 in California, and were awarded \$582,000 on average, and companies also have to spend from \$50,000 to \$250,000 to defend these cases (Oppenheimer 2003). Dertouzos et al. (1988) report that of the 120 wrongful discharge cases in California, plaintiffs won in 68% of the trials, with an average award of \$650,000, or about 18 years' salary. Similar large-amount awards are found true also in other states (e.g., Jung 1997, Abraham 1998).

Companies respond on two margins to minimize the costs of WDL-enabled litigation. On the one hand, they take costly measures such as severance pay, extended benefits, and outplacement assistance in the process of layoffs (e.g., Feldman and Leana 1989). On the other hand, they purchase employment practices liability

Figure 1. (Color online) Number of States That Adopted Wrongful Discharge Laws Over Time



Notes. This figure shows the number of states that have adopted the good faith, implied contract, and public policy exceptions to the employment-at-will doctrine over the 1970–2003 period. The vertical reference line is 1980, the starting year of our sample.

insurance (EPLI), which gained popularity in 1990s following the increase in wrongful discharge claims, to offset losses from these claims (e.g., Klenk 1999, Davis and Gassman 2013, Meyers and Hersch 2020).⁴ Overall, WDLs are found to have led to increased labor costs (Bird and Knopf 2009).

Real effects of WDLs are also examined both at the economy level and at the firm level. For instance, Dertouzos and Karoly (1992) and Autor et al. (2004, 2006) document a decline in overall employment in states that adopt WDLs; in contrast, Miles (2000) finds no difference in employment across states with and without WDLs, and MacLeod and Nakavachara (2007) show WDLs lead to increased employment. Similarly inconclusive is their effect on firm creation. Using establishment-level Annual Survey of Manufacturers data from the U.S. Census Bureau, Autor et al. (2007) show slower employment flows and firm entry rates, as well as an overall decline in total factor productivity associated with WDLs. Also using establishment-level data from the U.S. Census Bureau yet of the Business Dynamics Statistics (BDS) database, Acharya et al. (2014) find the WDLs spur innovation as measured by public firms' patents as well as new firm creation. Consistent with the increased labor costs, Serfling (2016) finds that firms' operating leverage and earning variability increase after the adoption of WDLs, and they respond by reducing debt ratios.

The good faith exception is viewed as the strongest form of WDL by legal scholars (Dertouzos and Karoly 1992, Kugler and Saint-Paul 2004, Serfling 2016). Miles (2000) and Autor et al. (2007) point out that firms mitigate the effect of the implied contract by including disclaimers in their employees' handbooks stating that employment contracts are at will. Usually defined to include only statutes, the public policy exception is the least binding to employers (Kugler and Saint-Paul 2004). Consistent with the above arguments, Acharya et al. (2014) and Serfling (2016) both find only the good faith exception has the strongest effect on corporate policy.

3. Data and Methodology

3.1. Sample Selection

Our venture capital data source is the Thomson Reuters VentureXpert database. For every portfolio firm, we record its location and primary Standard Industrial Classification code, based on which we can assign it to certain state and sector. The same firm may receive multiple rounds of investment, and we treat each round of financing as a VC deal and record the date and total amount of capital raised in the round. We restrict attention to U.S.-based firms and screen out the following transactions: (1) private investments in public equity, leveraged buyouts, and others classified as "public

market" transactions; (2) transactions made by funds of funds; (3) financial, banking, insurance, and real estate companies; (4) utilities firms; (5) companies incorporated in the District of Columbia; and (6) companies that are 15 years old or older. We also manually excluded several misclassified buyout and merger and acquisition transactions. We transform the dollar amount of a VC deal by the Consumer Price Index (CPI) to the constant dollar of year 2000, and then aggregate the number, and the dollar amount, of VC deals in each state and year.

Our sample period is 1980–2003. The end date is governed by the fact that the last WDL in the United States was passed in 1998. We extend the sample five years beyond this point to capture any subsequent effects. In robustness checks, we extend the sample to 2008 to examine long-term effects. The year 1980 serves as the starting point because venture capital emerged as a significant player in the capital market only after changes were made to the "prudent man" rule of the Employee Retirement Income Security Act (ERISA) in 1979. These modifications allowed pension funds to invest in the VC industry (Gompers 1994, Kortum and Lerner 2000).

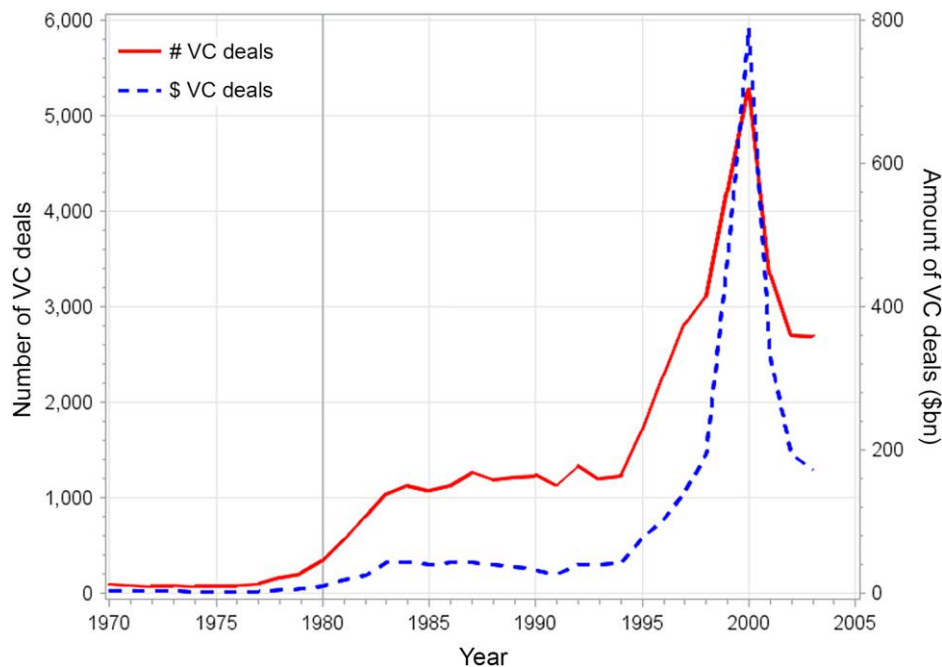
Our data set comprises 43,974 rounds of finance, encompassing 14,496 unique firms across 49 states (excluding Alaska, where no venture capital deals were identified). Figure 2 illustrates a surge in venture activity during the 1980–2003 period. Before the ERISA changes in 1979, the number of deals remained below 100. The VC market continued to evolve and grow alongside the commercialization of the internet in the 1990s, reaching a peak of 5,280 in 2000, during the dot-com bubble.

Figure 3 highlights the significant variations in VC investment observed across different states. California, for instance, recorded over 10,000 VC deals, whereas several states including Alaska, Louisiana, and West Virginia all registered fewer than 100 deals in our sample period.

3.2. State Characteristics

Cross-state differences in legal, economic, political, demographic, and educational factors might have a bearing on VC investment. Following the literature, we compile a set of state characteristic data from various sources. The first state policy to consider is unemployment insurance, which provides a buffer to workers who are fired. Unemployment insurance is based on the archived "Significant Provisions of State UI Law" files available from the U.S. Department of Labor website (<https://oui.doleta.gov/unemploy/statelaws.asp>) and measured as the maximum benefit possible, that is, the product of the maximum weekly benefit and the maximum duration allowed (Agrawal and Matsa 2013). Noncompete agreements prevent workers from working for "competitor" companies during or after their current employment, which might disincentivize employee effort to innovate

Figure 2. (Color online) VC Activities in the United States Over Time

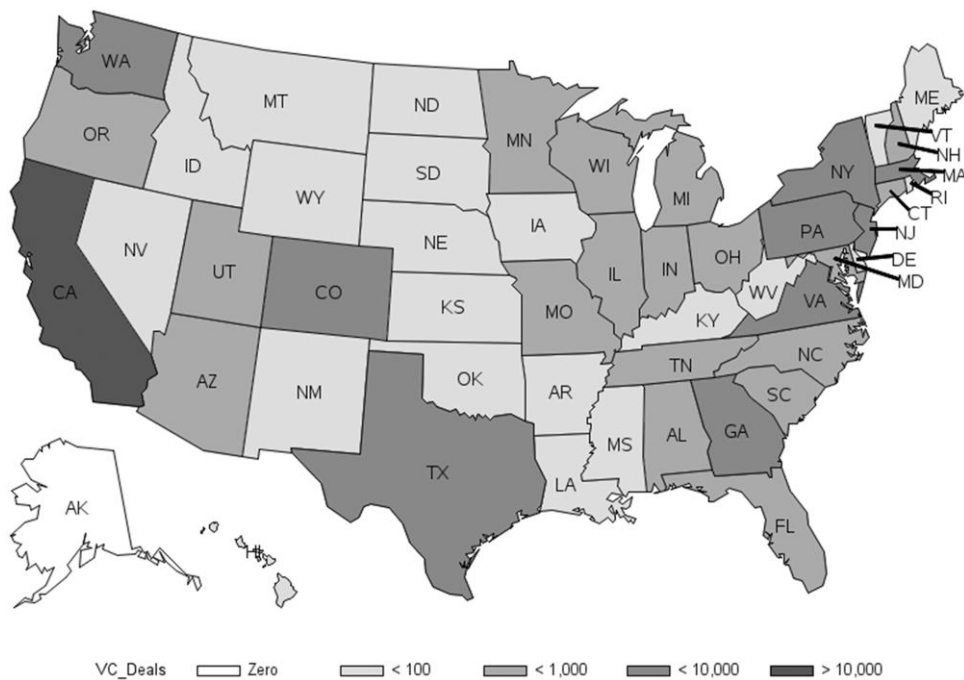


Notes. This figure shows the number and dollar amount of VC investment deals in the United States during 1970–2003. Amounts of VC deals are in constant year 2000 dollars. The vertical reference line is 1980, the starting year of our sample.

(Fulghieri and Sevilir 2011). We control for state-level noncompete laws by including the noncompete enforceability index (Garmaise 2012, Bird and Knopf 2015) as an additional control variable.⁵ More entrepreneur-friendly bankruptcy laws encourage more vibrant venture capital

activities. More entrepreneur-friendly personal bankruptcy laws encourage more vibrant entrepreneurial activities (Fan and White 2003, Armour and Cumming 2008, Cumming and Li 2013). Following the literature, we use the homestead exemption to measure the rigidity

Figure 3. VC Activities Across States



Note. This figure shows the frequency of venture capital investment deals in 50 U.S. states during 1980–2003.

of state-level bankruptcy laws.⁶ Smith et al. (2022) show that the Freedom of Information Act (FOIA) may reduce bias in VC reporting of their performance. We obtain the FOIA enactment year at the state level from Wikipedia (2024) and set a dummy variable, *FOIA*, equal to one after the enactment year and zero otherwise.

Population, real gross domestic product (GDP), and real GDP growth rate are extracted from the U.S. Bureau of Economic Analysis database. We take the logarithms of both population and real GDP and express the real GDP growth rate as a percentage. The Small Business Innovation Research (SBIR) program awards contracts or grants to small businesses nationwide and is shown to enhance recipient companies' chances of getting VC funding in later years (Lerner 1999, Cumming and Li 2013). We aggregate the total dollar amount of SBIR awards for each state in each year, adjust it by the CPI, and take the logarithm. The development of higher education in a state may contain information about the technological and intellectual potential. We thus collect the number of students enrolled in degree-seeking higher education institutions in each state and year from the Annual Statistical Abstracts of the U.S. Census Bureau. We lag this variable by five years to better gauge the size of the candidate pool of first-time entrepreneurs. From the same source, we also obtain, separately, the numbers of Democratic and Republican representatives in the House of Representatives in each state and year, and then use the Democrat-to-Republican ratio to measure the political balance. We control for political balance because it has implications on a wide array of economic and business issues such as taxation and entrepreneurship (e.g., Peters 1991, Beland and Unel 2019). Chircop et al. (2020) find that VCs located in more religious counties make less risky investments. We measure religiosity as the total number adherents to any recognized religious denomination scaled by the population in each state-year, based on data from the Association of Religion Data Archives' (ARDA) decennial surveys.⁷

We also estimate a hazard model to see whether the adoption of WDLs can be explained by the prior VC investment. This model employs two additional sets of explanatory variables. The first set is the fractions of states in the same federal circuit region that have adopted the good faith, implied contract, and public policy exceptions, respectively, in a given year. The second set of variables is the union membership density (measured as the fraction of nonagricultural employees in a state who are union members in a given year), the state unemployment rate, and their changes from the preceding year. These variables are deemed relevant because the decline in unionization and increase in unemployment are among primary motivations of the WDLs (Walsh and Schwarz 1996).

State judges routinely consider whether other states have adopted a similar exception when making their own WDL-related rules (Walsh and Schwarz 1996, Bird and Smythe 2008).

For detailed variable definitions, please see the appendix.

3.3. Summary Statistics

Table 1 presents the summary statistics of variables in our sample during 1980–2003. There are 1,200 state-year observations over the 24-year period. On average, a state has about 37 VC deals in a year, with a total dollar value of \$249 million. VC activity varies vastly across state-years: some states had no VC investment at all in certain years, whereas the maximum number (value) of VC deals in a state-year is 2,020 (\$33 billion). In 17% of the state-years, the good faith exception has been adopted, whereas the implied contract exception is in place in 67% of observations, and the public policy exception in 70% of observations. The maximum amount of unemployment insurance averages \$6,259 and spans a wide range from about \$2,340 to over \$23,000. Thus, variations abound in both labor market variables and VC investment.

Population and GDP also vary across states. The real state GDP growth rate averages 2.64%. The maximum benefit of unemployment insurance ranges from \$2,340 to \$23,040, whereas homestead exemptions for personal bankruptcy range from zero to unlimited.⁸ Political balance, the ratio of Democrat-to-Republican representatives in the state House of Representatives, ranges from 0 to 11. Based on the ARDA data, 52% of an average state's population adheres to a certain religion, consistent with the literature. The Freedom of Information Act is enacted in 89% of the state-years. The noncompete enforceability index ranges from zero to nine with a median of five. The average SBIR award received by small businesses in a state is about \$14 million a year. The median value is much lower, at about \$3 million, and the maximum value is \$373 million. On average, higher education institutions in a state enroll about 275,000 students.

Circuit WDLs have averages consistent with the good faith, implied contract, and public policy indicators in the sample, but with smaller standard deviations. Across all observations, 15.2% of workers are unionized, and unemployment rate averages 6%. Changes in these two metrics are slightly negative, reflecting the downward trend in unionization and unemployment over the sample period.

3.4. Empirical Methodology

The staggered adoption of WDLs enables us to conduct difference-in-differences analyses of their impact on VC investment. We follow the same general empirical framework as Autor et al. (2006), Acharya et al. (2014),

Table 1. Summary Statistics

Variable	N	Mean	Median	Std. dev.	Min	Max
Number of VC deals	1,200	36.6	6	125.3	0	2,020
Amount of VC deals (\$millions)	1,200	249.0	20.5	1,362.7	0	33,335
Good Faith	1,200	0.17	0	0.38	0	1
Implied Contract	1,200	0.67	1	0.47	0	1
Public Policy	1,200	0.70	1	0.46	0	1
Population (millions)	1,200	5.09	3.46	5.53	0.41	35.25
GDP (\$billions)	1,200	128.9	69.7	172.0	4.9	1,527.3
Real GDP growth rate (%)	1,200	2.64	2.60	3.91	−27.44	31.17
Unemployment insurance (\$)	1,200	6,259	5,824	2,464	2,340	23,040
Homestead exemptions	1,200	11.02	10.40	5.67	0	28.33
Political balance	1,200	2.40	1.02	3.42	0	11.00
Religiosity	1,200	0.52	0.54	0.12	0.29	0.80
FOIA	1,200	0.89	1	0.31	0	1
Noncompete index	1,200	4.25	5	1.78	0	9
SBIR (\$millions)	1,200	14.13	2.94	33.13	0	372.84
College enrollment (millions)	1,200	0.28	0.18	0.33	0.02	2.47
Circuit Good Faith	1,200	0.17	0	0.22	0	0.67
Circuit Implied Contract	1,200	0.67	0.80	0.33	0	1
Circuit Public Policy	1,200	0.70	0.86	0.35	0	1
Union membership density	1,200	0.15	0.14	0.07	0.03	0.38
Unemployment rate	1,200	0.060	0.056	0.021	0.023	0.178
Δ Union membership density	1,150	−0.004	−0.004	0.014	−0.065	0.072
Δ Unemployment rate	1,150	0.000	−0.001	0.010	−0.036	0.039

Notes. This table presents the summary statistics of variables at the state level during the 1980–2003 sample period. Variable definitions are in the appendix. Std. dev., Standard deviation.

Serfling (2016), and Bai et al. (2020). This approach views states that adopted WDLs during our sample period as treated observations, and states without WDLs throughout the whole sample period and states that had WDLs in place at the beginning of the sample period as the control group. We compare the post- and pre-WDL differences across the treatment group and the control group. The difference-in-differences captures the net effect of WDLs on VC investment.

Specifically, we estimate the following TWFE panel model:

$$VC_{st} = \beta_1 GF_{st} + \beta_2 IC_{st} + \beta_3 PP_{st} + \beta_4 X_{st} + \gamma_s + \delta_t + \varepsilon_{st}, \quad (1)$$

where VC_{st} is the VC investment in state s and year t ; GF_{st} , IC_{st} , and PP_{st} are dummy variables to indicate whether state s has adopted the good faith, implied contract, and public policy exceptions, respectively, as of year t ; and X_{st} is a set of state characteristics brought in as control variables, including population, real GDP, real GDP growth rate, unemployment insurance, homestead exemptions, political balance, religiosity, an FOIA indicator, noncompete index, SBIR awards, and college enrollment. The model also includes state fixed effects, γ_s , and year fixed effects, δ_t . The state fixed effects control for unobservable heterogeneity across states, and the year fixed effects account for intertemporal technological shocks as well as other economic, regulatory, and legal changes that potentially impact the whole VC market. Following prior literature, we cluster standard

errors at the state level to account for potential serial correlation in the data (e.g., Bertrand and Mullainathan 2003, Acharya et al. 2014).⁹

In this model, our primary focus lies on the effects of the good faith, implied contract, and public policy exceptions, denoted by β_1 , β_2 , and β_3 , respectively, on VC investment. Our hypothesis postulates a negative relationship between WDLs and VC investment, with the good faith exception having the most substantial impact. This assertion aligns with previous research findings, which consistently highlight the significant influence of the good faith exception compared with other WDLs (Acharya et al. 2014, Serfling 2016, Bai et al. 2020). We therefore hypothesize negative coefficients for these variables, with particular emphasis on the indicator for the good faith exception.

4. Main Results

4.1. Baseline Regressions

Table 2 presents the baseline difference-in-differences results based on ordinary least squares (OLS) estimation. VC investment is measured by the number, and dollar amount, of VC deals in a state-year. Because about 20% of state-years have zero VC investment, we take the logarithm of the sum of VC investment and one for most of our regressions to mitigate the severe skewness of the data. For each dependent variable, we employ two specifications: specification (1) has only employment protection variables, namely, the *Good Faith*, *Implied Contract*, and *Public Policy* indicators as

Table 2. WDLs and State-Level VC Investment

Independent variables	N(VC deals)		\$(VC deals)	
	(1)	(2)	(1)	(2)
<i>Good Faith</i>	−0.254*** (0.074)	−0.249*** (0.078)	−0.477*** (0.125)	−0.437*** (0.134)
<i>Implied Contract</i>	−0.130** (0.056)	−0.065 (0.055)	−0.231** (0.095)	−0.143 (0.094)
<i>Public Policy</i>	0.036 (0.057)	0.047 (0.057)	0.130 (0.097)	0.117 (0.097)
<i>State population</i>		0.039 (0.345)		−0.486 (0.590)
<i>Real state GDP</i>		1.012*** (0.201)		1.688*** (0.343)
<i>Real GDP growth rate</i>		0.289 (0.465)		0.492 (0.794)
<i>Unemployment insurance</i>		0.237 (0.172)		0.169 (0.295)
<i>Homestead exemptions</i>		−0.006 (0.012)		−0.021 (0.020)
<i>Political balance</i>		−0.056** (0.028)		−0.150*** (0.048)
<i>Religiosity</i>		−0.153 (0.699)		−2.101* (1.193)
<i>Economic freedom</i>		0.015 (0.076)		0.122 (0.130)
<i>Noncompete index</i>		−0.054* (0.030)		−0.081 (0.051)
<i>Government VC</i>		−0.005 (0.007)		−0.005 (0.012)
<i>College enrollment</i>		0.959*** (0.314)		1.324** (0.535)
State fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
F-test for no fixed effects: $Pr > F$	<0.001	<0.001	<0.001	<0.001
Observations	1,200	1,200	1,200	1,200
R ²	0.492	0.537	0.555	0.589

Notes. This table presents the difference-in-differences estimates of the effects of good faith, implied contract, and public policy exceptions to at-will employment on state-level VC investment in the 1980–2003 period. The dependent variables are $\ln(1 + \text{Number of VC deals})$ and $\ln(1 + \text{Dollar amount of VC deals})$ in a state-year. *Good Faith* is an indicator variable set to one if the state has passed the good faith exception by the year and zero otherwise; *Implied Contract* and *Public Policy* are indicators defined similarly. State characteristics, as defined as in Table 1, are controlled for in specification (2). State and year fixed effects are controlled for in all specifications. Standard errors are clustered at the state level and reported in parentheses.

*, **, and *** mark statistical significance at the 10%, 5%, and 1% levels, respectively.

explicit explanatory variables, and specification (2) brings in state characteristics as control variables. State and year fixed effects are controlled for in both specifications.

In all four models, the *Good Faith* indicator consistently yields negative and statistically significant coefficients. The magnitude of these coefficients is large when compared with those associated with the *Implied Contract* and *Public Policy* indicators. Regardless of explicitly controlling for state characteristics, the coefficient is about −0.250 when the dependent variable is the count

of VC deals, indicating that states on average experience a decline of approximately 22% in the number of VC deals after adopting the good faith exception.¹⁰ Likewise, the coefficient in the last column is −0.427, representing approximately a 35% decline in total dollar amount of VC investment. These numbers are economically large.

The *Implied Contract* indicator also exhibits negative coefficients in all four models, but these coefficients are generally small, and statistically insignificant when controlling for state characteristics. The coefficients on the

public policy exception are never significant. To summarize, out of the three WDL exceptions, the good faith exception exhibits a strong negative effect on VC investment, the implied contract exception has a very weak negative effect, and the public policy exception does not appear to have an impact. This echoes the literature that the good faith exception is the most stringent exception.

Table 2 also illustrates which state characteristics affect VC investment. First, real GDP exhibits positive and statistically significant coefficients, indicating that a well-developed economy is associated with increased VC activity. On the other hand, both unemployment insurance and homestead exemptions fail to yield coefficients that significantly differ from zero, suggesting no substantial impact on VC investment. Political balance demonstrates a negative coefficient, indicating that an increase in the proportion of Democratic House representatives relative to Republican representatives is associated with less VC activity. Religiosity is associated with negative coefficients, with marginal statistical significance observed when the dependent variable is the dollar amount. This finding aligns with existing literature suggesting that individuals with higher levels of religious adherence are less inclined to engage in risky investments. Noncompete laws display negative effects on VC investment, potentially attributable to their discouraging impact on innovation. As anticipated, past college enrollment exhibits a positive effect on VC investment, as recent college graduates constitute a significant pool of potential entrepreneurs.

Overall, these findings shed light on the influence of state characteristics on VC investment, highlighting the importance of factors such as economic development, political balance, religiosity, noncompete laws, and educational attainment in shaping the VC landscape.

4.2. Robustness Check

4.2.1. Alternative Samples. The last WDL in our sample was passed in 1998. Our baseline sample period extends to 2003 in order to observe up to five years of subsequent effects of the law's passage. There are trade-offs involved in setting this horizon, owing to the serial correlation problem associated with long sample period (Bertrand et al. 2004). In a robustness test, we further extend our sample period to 2008, allowing 10 years after the last adoption of the good faith exception. The results with state characteristics controlled for are reported in the left section of Table 3, panel A. The *Good Faith* indicator loads negatively regardless of the measure of VC investment; the *Implied Contract* indicator also receives negative coefficients, but they are not statistically significant; and the *Public Policy* indicator's coefficients have mixed signs and are not statistically different from zero. These results are qualitatively consistent with the baseline results in Table 2.

Our sample period also contains one highly unusual event: the dot-com boom and its subsequent crash. To mitigate the potential bias associated with the outlier observations in this period, we exclude years 2000 and 2001 from the sample and reestimate our model. As displayed in the middle section of Table 3, panel A, the coefficients on the *Good Faith* indicator do not change qualitatively. The *Implied Contract (Public Policy)* indicator receives a negative (positive) and marginally significant coefficient when the VC investment is measured by the dollar amount. Overall, the results are similar with or without the dot-com peak and crash included.

Several states warrant further inspection. California and Massachusetts are hotspots of VC investment, contributing about half of all VC deals during our sample period. As one of the few states that adopted all the three WDL exceptions, Alaska, in contrast, has no VC deals at all during our sample period. To purge the effect of these outliers, we rerun our tests in a sample excluding California and Massachusetts, and in another excluding Alaska. We report the estimates in the rightmost side of panel A and the leftmost side of panel B, respectively, in Table 3. The results are qualitatively the same as the baseline results in Table 2.

4.2.2. Alternative Log Transformation. Additionally, we use $\ln(\text{VC investment})$ to replace $\ln(1 + \text{VC investment})$ as the dependent variable. This new transformation excludes 231, or 19.3% of the state-year observations with zero VC investment, and we are left with 969 observations in the sample. The DiD results are presented in the middle section of Table 3, panel B. When the dependent variable is based on count of VC deals, the results are qualitatively identical to those in Table 3, that is, the *Good Faith* indicator loads negatively when the *Implied Contract* and *Public Policy* indicators receive negative and positive, respectively, coefficients that are not statistically significant. When the dependent variable is based on the dollar amount of VC deals, the *Good Faith* indicator continues to load negatively; the *Implied Contract* loads negatively, too.

Under the alternative logarithm transformation, the results are qualitatively similar but with slightly larger coefficients on the good faith exception. A possible explanation is that this transformation disproportionately excludes observations in states with lower intertemporal variations in VC investment, which may have upwardly biased the apparent effects of WDLs.

4.2.3. Alternative Estimator. Recent literature highlights certain challenges associated with OLS estimation when applied to the two-way fixed effect model. In our specific case, the two-way fixed effects specification entails two types of comparisons. First, it involves comparing the treated units with the untreated units, and second, it includes comparisons between the treated

Table 3. Robustness Tests: Alternative Samples, Dependent Variables, and Estimator

Panel A. Alternative samples						
Independent variables	Sample period 1980–2008		Excluding bubble period		Excluding CA and MA	
	<i>N</i> (VC deals)	\$(VC deals)	<i>N</i> (VC deals)	\$(VC deals)	<i>N</i> (VC deals)	\$(VC deals)
<i>Good Faith</i>	−0.193*** (0.072)	−0.376*** (0.120)	−0.236*** (0.081)	−0.432*** (0.136)	−0.262*** (0.081)	−0.461*** (0.138)
<i>Implied Contract</i>	−0.069 (0.053)	−0.131 (0.089)	−0.075 (0.055)	−0.161* (0.092)	−0.062 (0.058)	−0.142 (0.098)
<i>Public Policy</i>	0.048 (0.055)	−0.100 (0.092)	0.077 (0.057)	0.157* (0.096)	0.047 (0.059)	0.118 (0.100)
Observations	1,450	1,450	1,100	1,100	1,152	1,152
Panel B. Alternative sample, different transformation of independent variables, and different estimator						
Independent variables	Excluding AK		Log(VC)		dC-D'H estimator	
	<i>N</i> (VC deals)	\$(VC deals)	<i>N</i> (VC deals)	\$(VC deals)	<i>N</i> (VC deals)	\$(VC deals)
<i>Good Faith</i>	−0.252*** (0.081)	−0.470*** (0.138)	−0.386*** (0.095)	−0.630*** (0.177)	−0.794*** (0.245)	−1.344* (0.730)
<i>Implied Contract</i>	−0.067 (0.056)	−0.152 (0.095)	−0.079 (0.063)	−0.275** (0.118)	−0.195 (0.264)	−0.322 (0.393)
<i>Public Policy</i>	0.045 (0.058)	0.109 (0.098)	0.061 (0.070)	0.182 (0.129)	0.033 (0.180)	0.171 (0.352)
Observations	1,176	1,176	969	969	1,200	1,200

Notes. Panel A of this table presents the DiD estimates of the effects of good faith, implied contract, and public policy exceptions to at-will employment on state-level VC investment in alternative samples, including the sample with the extended period 1980–2008, the sample with the dot-com bubble period (2000–2001) excluded, and the sample excluding California and Massachusetts, the two states with the most VC investments. In panel B, we first use the sample excluding states that have the good faith exception in place throughout the whole sample period (“early bird” states). We also employ the de Chaisemartin and D’Haultfoeuille (2022) estimator to address the econometric concerns of the traditional DiD estimator with staggered events. The coefficients on the three WDL exceptions are estimated separately controlling for the other two treatments. For all the above robustness tests, the dependent variables are $\ln(1 + \text{Number of VC deals})$ and $\ln(1 + \text{Dollar amount of VC deals})$ in a state-year. Last, we remove the 231 state-years with zero VC deals and use $\ln(\text{Number of VC deals})$ and $\ln(\text{Dollar amount of VC deals})$ as the dependent variables in the rightmost side of panel B. *Good Faith* is an indicator variable set to one if the state has passed the good faith exception by the year and zero otherwise; *Implied Contract* and *Public Policy* are indicators defined similarly. State characteristics and state and year fixed effects are controlled for in all specifications. Standard errors are clustered at the state level and reported in parentheses. AK refers to Alaska.

*, **, and *** mark statistical significance at the 10%, 5%, and 1% levels, respectively.

units and the pretreated units, often referred to as “forbidden comparisons” (Borusyak and Jaravel 2024). De Chaisemartin and D’Haultfoeuille (2020) and Goodman-Bacon (2021) demonstrate that the estimated effect in a two-way fixed effects model represents a weighted average of the true treatment effect, wherein the weights may turn out to be negative because of these forbidden comparisons. Consequently, even if the treatment effect is genuinely positive for all treated units, the estimated effect may erroneously appear negative.

The literature proposes several new estimators to address the above issue.¹¹ We employ the estimator from de Chaisemartin and D’Haultfoeuille (2020, 2022) (hereafter referred to as the dC-D’H estimator), as it allows for treatment reversals given the reversals of the good faith exception in New Hampshire and Oklahoma and of the implied contract exception in Arizona and Missouri in our sample. The dC-D’H estimator generates the estimates for each WDL exception separately while controlling for the other two exceptions and state

characteristics, and we report the estimates in the rightmost section of Table 3, panel B.¹² The *Good Faith* indicator obtains negative coefficients that are statistically significant, especially when VC investment is measured by the number of deals. The *Implied Contract* indicator receives negative coefficients, too, but they are not statistically different from zero. The coefficients on the *Public Policy* indicator are positive but indistinguishable from zero. Qualitatively, these results are consistent with baseline OLS estimates.

4.3. Dynamic Effects of WDLs

In this section, we examine the dynamics of VC investment around the adoption of WDLs. This practice helps mitigate the concern of potential reverse causality as well as provide the dynamics of the WDL effect (Bertrand and Mullainathan 2003). Specifically, for the good faith exception, we construct four timing indicators: Good Faith^{-1} , Good Faith^0 , Good Faith^1 , and Good Faith^{2+} , where the superscripts denote the year relative to the

adoption of the good faith exception in a given state. Thus, $Good\ Faith^{-1}$ is equal to one if a state adopts the exception in the following year and zero otherwise; $Good\ Faith^0$ is equal to one if a state adopts the exception in the current year; $Good\ Faith^1$ and $Good\ Faith^{2+}$ are equal to one if a state adopted the exception one year ago, and two or more years ago, respectively. These four indicators replace the good faith indicator in the baseline model. We keep the implied contract and public policy indicators unchanged because they have been found not impactful in previous tests.¹³ Then we bring in the state characteristics as well as state and year fixed effects, estimate the model using OLS, and report the estimated coefficients of WDL indicator variables in Table 4.

The indicator $Good\ Faith^{-1}$ receives positive coefficients that are statistically not different from zero in all the four specifications. In contrast, $Good\ Faith^0$, $Good\ Faith^1$, and $Good\ Faith^{2+}$ all obtain negative coefficients of much larger magnitudes; those on $Good\ Faith^0$ are statistically significant when the dependent variable is count of VC deals; those on $Good\ Faith^1$ and $Good\ Faith^{2+}$ load negatively in all four specifications, and the statistical significance is strong especially for the latter. This indicates that VC investment decline takes place only after the adoption of the good faith exception, lasting far

into post-good faith years. This pattern is inconsistent with VC activity leading to the WDLs.

4.4. Survival Analysis

Section 4.3 shows that changes in VC investment do not precede the adoption of WDLs. Walsh and Schwarz (1996) also point out that rationales for states to adopt WDLs are largely orthogonal to an objective to promoting VC investment or economic growth at large. Instead, state courts focused on fairness in employment relationships and consistency with established principles of contract law. Nevertheless, we conduct a survival analysis of the WDL adoptions to alleviate the residual concern of endogeneity, in a similar vein as Acharya et al. (2014) and Serfling (2016).

We utilize the Cox proportional hazard model to investigate the adoption of a WDL as a failure events. This framework enables us to identify the factors that influence WDL adoption. Each state remains in the sample until the year it adopts the exception. The primary independent variable of interest is VC investment, measured by the number or dollar amount of VC deals. To account for potential correlations among the exceptions, we include the other two exceptions as independent variables.

Table 4. Dynamic Effects of Good Faith Adoption on VC Investment

Independent variables	Ln(Number of VC deals)		Ln(Dollar Amount of VC Deals)	
	(1)	(2)	(1)	(2)
$Good\ Faith^{-1}$	0.075 (0.166)	0.078 (0.160)	0.108 (0.282)	0.089 (0.274)
$Good\ Faith^0$	−0.315** (0.154)	−0.256* (0.150)	−0.383 (0.262)	−0.311 (0.256)
$Good\ Faith^1$	−0.423*** (0.156)	−0.383** (0.152)	−0.535** (0.264)	−0.501* (0.259)
$Good\ Faith^{2+}$	−0.366*** (0.086)	−0.336*** (0.094)	−0.635*** (0.147)	−0.579*** (0.161)
<i>Implied Contract</i>	−0.146*** (0.056)	−0.081 (0.055)	−0.258*** (0.095)	−0.169* (0.094)
<i>Public Policy</i>	0.063 (0.058)	0.065 (0.057)	0.170* (0.098)	0.144 (0.098)
Control variables	No	Yes	No	Yes
State fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
F-test for fixed effects: $Pr > F$	<0.001	<0.001	<0.001	<0.001
Observations	1,200	1,200	1,200	1,200
R ²	0.497	0.540	0.558	0.591

Notes. This table presents the results from OLS regressions relating the timing of state-level VC investment to the enactment of good faith and implied contract exceptions to at-will employment during 1980–2003. The dependent variables are $\ln(1 + \text{Number of VC deals})$ and $\ln(1 + \text{Dollar amount of VC deals})$ in a state-year. $Good\ Faith^{-1}$ (*Implied Contract*^{−1}) is an indicator variable set to one if a state will adopt the GF exception (IC exception) in one year and zero otherwise; $Good\ Faith^0$ (*Implied Contract*⁰) is an indicator variable set to one if a state adopts the GF (IC) exception in the current year and zero otherwise; $Good\ Faith^1$ (*Implied Contract*¹) is an indicator variable set to one if a state adopts the GF (IC) exception in the preceding year and zero otherwise; $Good\ Faith^{2+}$ (*Implied Contract*²⁺) is an indicator variable set to one if a state adopts the GF (IC) exception two or more years ago and zero otherwise; and *Public Policy* is an indicator that is equal to one if the state has adopted the public policy exception in year t , defined similarly. Standard errors are clustered at the state level and reported in parentheses.

*, **, and *** mark statistical significance at the 10%, 5%, and 1% levels, respectively.

We control for state population, real GDP, and real GDP growth. We also incorporate unemployment insurance, which serves as another form of labor protection. In line with Serfling (2016), we additionally introduce two sets of control variables. The first set considers the fraction of states within the same federal circuit that have adopted a similar exception. The second set encompasses the level and year-over-year change of union membership density and unemployment rate. All independent variables are lagged in the analysis.

Estimation results are reported in Table 5. Notable observations are the following. First of all, neither VC count nor VC amount loads, regardless of the failure event. Thus, the adoption of WDLs is not related to past VC investment, further alleviating the concern of reverse causality. Second, unemployment insurance consistently obtains negative and statistically significant coefficients. This is in accordance with the concept that the lack of adequate unemployment protection is a trigger for enhanced employment protection. Third, the

Table 5. Hazard Model of Adoption of WDL Exceptions

Independent variables	Good Faith		Implied Contract		Public policy	
	(1)	(2)	(1)	(2)	(1)	(2)
<i>N(VC deals)</i>	0.481 (0.509)		0.199 (0.244)		0.041 (0.287)	
<i>\$(VC deals)</i>		0.093 (0.322)		0.022 (0.174)		−0.032 (0.174)
<i>Good Faith</i>			−0.021 (0.726)	0.107 (0.710)	1.821*** (0.651)	1.825*** (0.662)
<i>Implied Contract</i>	−0.105 (0.843)	−0.149 (0.845)			0.106 (0.467)	0.096 (0.482)
<i>Public Policy</i>	−0.816 (1.154)	−0.903 (1.189)	0.318 (0.475)	0.299 (0.470)		
<i>Circuit Good Faith</i>	3.712 (2.921)	4.087 (3.029)				
<i>Circuit Implied Contract</i>			−0.264 (1.070)	−0.446 (1.104)		
<i>Circuit Public Policy</i>					1.800 (1.413)	1.756 (1.438)
<i>Population</i>	−3.847** (1.551)	−3.690** (1.591)	−1.137 (1.381)	−0.959 (1.438)	1.468 (0.101)	1.539 (1.025)
<i>GDP</i>	2.790 (1.793)	3.003 (1.906)	0.886 (1.380)	0.867 (1.453)	−1.949* (1.067)	−1.947* (1.070)
<i>Real GDP growth rate</i>	−0.110 (5.518)	0.187 (5.409)	−8.412* (4.503)	−7.442* (4.331)	−1.140 (4.563)	−0.688 (4.711)
<i>Unemployment insurance</i>	−6.601** (2.520)	−6.298** (2.524)	−5.182*** (0.937)	−4.978*** (0.931)	−4.166*** (0.966)	−4.053*** (1.007)
<i>Union membership density</i>	4.922 (7.668)	3.609 (7.954)	15.90*** (3.422)	15.91*** (3.432)	13.50*** (4.297)	13.61*** (4.312)
<i>Unemployment rate</i>	17.10 (13.72)	14.42 (13.23)	−3.476 (9.94)	−5.847 (10.30)	−0.486 (11.21)	−1.325 (11.44)
<i>ΔUnion membership density</i>	−19.97 (18.92)	−18.48 (19.03)	−0.618 (10.16)	−0.105 (10.35)	3.025 (11.35)	3.380 (11.32)
<i>ΔUnemployment rate</i>	30.78 (21.39)	28.98 (22.09)	30.18** (14.83)	29.36** (14.65)	3.272 (16.45)	2.347 (16.78)
Observations	960	960	367	367	359	359
Pseudo- R^2	0.040	0.041	0.191	0.190	0.141	0.141
Likelihood ratio	39.16	39.9244	77.8224	77.1806	54.6081	54.6148

Notes. This table reports the estimation coefficients of Cox proportional hazard models in which the “failure events” are the adoption of WDLs, including the good faith, implied contract, and public policy exceptions, respectively, in a given U.S. state during the 1980–2003 period. For each exception, states are dropped from the sample once they pass the exception. Explanatory variables including VC investment in a state, state characteristics, other WDL exceptions, and the WDL adoption rate in the circuit, all measured as of year $t - 1$. Variable definitions are in Table 1. Standard errors are clustered at the state level and reported in parentheses.

*, **, and *** mark statistical significance at the 10%, 5%, and 1% levels, respectively.

passage of good faith exception is not influenced by the adoption of the other two exceptions, but it increases the likelihood the public policy exception is subsequently enacted.

4.5. Randomized WDL Adoption

VC investment neither precedes nor predicts the adoption of WDLs. The next question is, can the connection between WDLs and subsequent VC investment be a coincidence? To address this concern, we randomly select states to adopt the WDL exceptions in randomly picked years and check whether a similar connection emerges. Specifically, we randomly assign 12 good faith adoption events, 43 implied contract events, and 41 public policy events to the 50 states during the 1970–2003 period, that is, the period when all but one WDL enactment decisions occurred in the real world. We then cut the period to 1980–2003 to create a counterfactual sample in which we run the baseline difference-in-differences tests and record the estimated coefficients on the *Good Faith*, *Implied Contract*, and *Public Policy* indicators. Table 6 summarizes these coefficients in 500 simulated counterfactual samples.

As Table 6 shows, all three indicators receive coefficients that have near-zero means and medians, regardless of the measure of VC investment and whether state characteristics are included in the model specification. Reported in the last two columns, the *t*-test and Wilcoxon signed rank test show the mean and median coefficients are indistinguishable from zero. We conclude that the WDL effect on VC investment is not random because it is not observed in any coefficient in our counterfactual samples.

4.6. New Firm Creation vs. VC-Firm Creation

Acharya et al. (2014) show that the passage of WDLs, particularly the good faith exception, leads to creation of more new firms. Their rationale is that WDLs enhance employees’ innovative efforts and lead to more generic innovations, which in turn leads to the increase in creation of new firms. Using the establishment-level BDS data and a difference-in-differences approach similar to ours, they find a significant positive relation between the adoption of the good faith exception and, among others, the number of establishments created by start-ups. This result seems at odds with our finding that WDLs depress VC investment.

We first show that these two disparate results—more overall firms created, yet fewer VC-backed firms created—coexist in our data set. We then examine the possible reasons.

We consider the VC funding rate, defined as the number of start-ups that receive VC funding, divided by the number of business establishments created in the preceding three-year period. Our justification for a three-year period is assuming that a suitable start-up might receive seed VC funding within three years of its founding date. We employ two variations of this measure for robustness. First, we use a five-year period. Second, we use the number of newly created firms rather than establishments. Data of establishments and firms in all the 50 states are extracted from the BDS database. New establishments (firms) are defined as establishments (firms) with age zero.

In Table 7, panel A, we replicate the Acharya et al. (2014) result, using the logarithm of number of new establishments in a state as the dependent variable and

Table 6. Falsification Test: Randomized Enactment of WDL Exceptions

Dependent variable	Model	Independent variable	N	Mean	Median	Std. error	Mean ≠ 0 <i>Pr</i> > <i>t</i>	Median ≠ 0 <i>Pr</i> > <i>S</i>
<i>N</i> (VC deals)	Without state characteristics	<i>GF</i>	500	0.010	0.008	0.008	0.246	0.298
		<i>IC</i>	500	0.004	0.003	0.005	0.404	0.325
		<i>PP</i>	500	0.002	0.006	0.005	0.721	0.652
	With state characteristics	<i>GF</i>	500	0.005	0.005	0.008	0.548	0.605
		<i>IC</i>	500	0.003	0.006	0.004	0.444	0.415
		<i>PP</i>	500	0.001	−0.001	0.004	0.832	0.786
<i>\$</i> (VC deals)	Without state characteristics	<i>GF</i>	500	0.012	0.015	0.014	0.400	0.395
		<i>IC</i>	500	0.005	0.006	0.007	0.492	0.498
		<i>PP</i>	500	−0.002	−0.007	0.007	0.812	0.686
	With state characteristics	<i>GF</i>	500	0.005	−0.005	0.013	0.674	0.870
		<i>IC</i>	500	0.004	0.006	0.007	0.570	0.433
		<i>PP</i>	500	−0.004	−0.011	0.007	0.563	0.465

Notes. This table presents the summary statistics of estimated coefficients of *Good Faith*, *Implied Contract*, and *Public Policy* in 500 falsification samples. To create a falsification sample, we randomly select 12 states as having adopted the good faith exception during the 1970–2003 period, and for each of these states randomly assign an enactment year. Likewise, we randomly assign an implied contract enactment year to 43 states and a public policy enactment year to 41 states. Then we define indicator variables *Good Faith*, *Implied Contract*, and *Public Policy* as equal to one after the enactment of the good faith, implied contract, and public policy exception, respectively. Then we restrict our sample period to 1980–2003. Difference-in-differences tests are conducted in each falsification sample to relate the enactment of these WDL exceptions to the number of VC deals and dollar amount of VC deals at the state level, and estimated coefficients of *GF*, *IC*, and *PP* are recorded for all the 500 falsification samples. The *t*-test and Wilcoxon signed rank test are used to test whether the mean and median, respectively, are statistically different to zero, and *Pr* > |*t*| and *Pr* > |*S*| are reported accordingly in the last two columns.

Table 7. WDL, Creation of New Firms, and VC Funding of New Firms

Panel A. New firm creation and VC funding rates						
Independent variables	Establishments	Firms	VC deals/Establishments		VC deals/Firms	
			3-year	5-year	3-year	5-year
<i>Good Faith</i>	0.053*** (0.013)	0.077*** (0.012)	−2.070** (0.823)	−1.272*** (0.483)	−2.090** (0.829)	−1.283*** (0.487)
<i>Implied Contract</i>	−0.012 (0.008)	−0.006 (0.008)	0.019 (0.578)	−0.099 (0.339)	0.022 (0.583)	−0.097 (0.342)
<i>Public Policy</i>	−0.014 (0.009)	−0.014 (0.009)	−0.201 (0.596)	−0.033 (0.349)	−0.208 (0.600)	−0.037 (0.353)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
State fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,200	1,200	1,200	1,200	1,200	1,200
R ²	0.718	0.712	0.493	0.491	0.493	0.490
Panel B. Firm creation by employee number						
Independent variables	Establishment size					
	[1, 4]	[5, 9]	[10, 19]	[20, 99]	[100, 499]	500+
<i>Good Faith</i>	0.657*** (0.197)	0.238*** (0.067)	0.047 (0.058)	0.036 (0.053)	−0.077 (0.099)	−0.146 (0.122)
<i>Implied Contract</i>	−0.099 (0.139)	−0.007 (0.046)	−0.017 (0.041)	−0.040 (0.038)	−0.115* (0.069)	0.166* (0.086)
<i>Public Policy</i>	−0.321** (0.143)	−0.038 (0.048)	0.063 (0.042)	0.035 (0.039)	0.007 (0.071)	−0.012 (0.089)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
State fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,200	1,200	1,200	1,200	1,200	1,200
R ²	0.719	0.678	0.606	0.557	0.228	0.110
Panel C. High-tech vs. non-high-tech firm creation						
Independent variables	Establishments		Entry		Entry rate	
	High-tech	Non-high-tech	High-tech	Non-high-tech	High-tech	Non-high-tech
<i>Good Faith</i>	0.013 (0.009)	0.031*** (0.004)	0.028 (0.021)	0.065*** (0.010)	0.215 (0.289)	0.486*** (0.111)
<i>Implied Contract</i>	−0.005 (0.007)	−0.001 (0.003)	−0.005 (0.015)	−0.008 (0.007)	0.098 (0.203)	−0.097 (0.078)
<i>Public Policy</i>	−0.016** (0.007)	0.005 (0.003)	−0.046*** (0.015)	−0.010 (0.007)	−0.626*** (0.209)	−0.181** (0.080)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
State fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,200	1,200	1,200	1,200	1,200	1,200
R ²	0.975	0.960	0.860	0.761	0.479	0.800

Notes. This table presents the DiD estimates of the effects of good faith, implied contract, and public policy exceptions to at-will employment on creation of new firms and the fraction of new firms that are funded by VC at the state level in the 1980–2003 period. In panel A, the dependent variables are *Establishments*, *Firms*, *VC deals/Establishments*, and *VC deals/Firms*, respectively. The latter two are referred to as VC funding rates and scaled by a factor of 10,000, that is, measured as basis points. *Establishments (Firms)* is measured as the logarithm of number of establishments (firms) of age zero in a state. *VC deals/Establishments* is the number of VC deals divided by the number of establishments created in the past three- or five-year period in a state. *VC deals/Firms* is the number of VC deals divided by the number of firms created in the preceding three- or five-year period in a state. These two VC funding rates are winsorized at the 99th percentiles. In panel B, the dependent variables are the log-transformed numbers of establishments with certain number of employees. In panel C, the dependent variables are the log-transformed numbers of establishments, new entries, and entry rate in high-tech or non-high-tech industries as identified in the BDS-High Tech database. *Good Faith* is an indicator variable set to one if the state has passed the good faith exception by the year and zero otherwise; *Implied Contract* and *Public Policy* are indicators defined similarly. State characteristics are controlled for. Standard errors are clustered at the state level and reported in parentheses.

*, **, and *** mark statistical significance at the 10%, 5%, and 1% levels, respectively.

the three WDL indicators as the main dependent variable. The *Good Faith* indicator obtains a coefficient of 0.053, statistically significant at the 1% level, whereas the coefficients on the *Implied Contract* and *Public Policy* indicators are essentially zero. This is consistent with the Acharya et al. (2014) finding that the good faith exception leads to an increase in new firm creation. When we measure firm creation by the number of new firms in the next column, the results are similar.

Then we replace the number of new establishments/firms by the four measures of VC funding rates and rerun the estimates. The *Good Faith* indicator obtains negative coefficients that are statistically significant in all the four specifications, indicating that after the adoption of the good faith exception, a smaller fraction of start-ups receives VC funding. This result helps to show why fewer VC deals are made when more firms are created. The *Implied Contract* and *Public Policy* indicators both obtain coefficients that are not statistically different from zero, implying that they do not affect the likelihood of start-ups receiving VC capital.

4.6.1. What Explains the Difference Between Creation Rates of VC-Backed Firms and Other Firms? To better understand the difference between creation rates of firm that receive VC deals and other firms evident in Table 7, panel A, consider that for anyone acting in the role of owner, WDLs are likely viewed in a negative light. Impaired operating flexibility reduces the value of the firm (and thus the net present value of the initial investment). VCs are clearly owners. WDLs should therefore reduce the flow of VC financial capital into the market. In contrast, for anyone in the role of employee, the effect of WDLs is positive for the reasons emphasized in Acharya et al. (2014). Entrepreneurs (or teams of entrepreneurs) are hybrids. They are literal employees of the firm, and as such are rightfully concerned about being fired by other owners. (Famously, the founders of Cisco were fired by their VCs just before the dot-com wealth frenzy.) On the other hand, entrepreneurs are also part owners who share in the upside enabled by operating flexibility.

The effect of WDLs on entrepreneurs is therefore ambiguous. It depends upon whether, for incentive purposes, they function more like employees or owners. Start-up creation will rise only when the positive impact on employee innovation more than compensates the decline in operational flexibility. Our results in Table 7, panel A, indicate this condition holds in some firms, but is strongly violated for the types of firms that VCs target. VCs invest in highly uncertain opportunities for which operating flexibility is the key source of value. For these types of firms, the costs of WDLs outweigh the benefits.

WDLs may increase overall firm creation through yet another channel. If WDLs make existing firms reluctant to hire new employees, then some of these individuals

will instead elect to go into self-employment.¹⁴ For example, in the field of accounting, the passage of WDLs may encourage a large number of self-employed accountants rather than accounts concentrated in a few large firms.

Although the census data lack a strict “self-employment” indicator, they group establishments into size buckets: one to four employees, five to nine employees, etc. The argument above suggests that following WDL passage, new firm creation should be concentrated in the smallest buckets. That is, we will not observe as many new large firms (creating them is prohibitively expensive because of firing costs), but instead observe small firms being created (including self-employment). Table 7, panel B, shows evidence consistent with this hypothesis. The *Good Faith* indicator is positive and statistically significant in the two smallest buckets, and the coefficients monotonically become smaller as one moves toward larger buckets.

To further examine the notion that WDLs lead to (i) more new firms overall but (ii) fewer new firms suitable for VC funding, we use the BDS-High Tech database to separate the establishments in the high-tech industries and those in non-high-tech industries (for details, see Goldschlag and Miranda 2020).¹⁵ We hypothesize that the firm creation seen in Table 7, panel A, is due to low-tech firms. We run model (1) separately for each subset of new establishments, and present the results in Table 7, panel C. The results are consistent with our hypothesis. The coefficients are positive and statistically significant for the good faith exception in the non-high-tech subset. The same coefficients in the high-tech subset are statistically insignificant.

5. VC Investment at the Sector Level

5.1. Baseline Investigation

It is well known that VC investment concentrates in a few sectors such as computer technology and biotechnology. Easy access to new ideas, talents and capital makes some locations hotspots of VC-backed start-ups of certain particular business sectors. For instance, computer-related start-ups flock in the Silicon Valley in California, whereas the Boston area is the birthplace of many biotech companies. This observation leads to a question: Do WDLs adopted by states influence VC investment at the sector level?

To answer this question, we modify model (1), as below, to accommodate difference-in-differences analyses at the state and sector level:

$$VC_{ist} = \beta_1 GF_{st} + \beta_2 IC_{st} + \beta_3 PP_{st} + \beta_4 X_{st} + \gamma_s + \delta_{it} + \varepsilon_{ist}, \quad (2)$$

where subscript i denote the i th business sector; VC_{ist} , then, is the VC investment in sector i , state s , and year t . The *Good Faith* (GF_{st}), *Implied Contract* (IC_{st}), and *Public*

Policy (PP_{st}) indicators remain unchanged, and so do state characteristics (X_{st}) and state fixed effects (γ_s). The term δ_{it} represents sector-year fixed effects to control for time trends within a sector, and ε_{ist} the error. Again, the coefficients of main interest are β_1 , β_2 , and β_3 .

We use the European Private Equity and Venture Capital Association (EVCA) standard to classify VC-backed firms into 15 business sectors (Bozkaya and Kerr 2014).¹⁶ For the 1980–2003 period, we have 18,000 observations at the state-sector level.

Table 8 reports the difference-in-differences results. In the baseline OLS estimation, the *Good Faith* indicator receives negative coefficients that are statistically significant at the 1% level in all specifications, regardless of the measure of VC investment, with or without controlling for state characteristics. Thus the good faith exception depresses VC investment at the sector level, corroborating our findings at the state level. Notably, the *Implied Contract* indicator also loads negatively in all the specifications, albeit with lower levels of magnitude and statistical significance. The *Public Policy* indicator receives coefficients that are not statistically significant. When we employ the dC-D'H estimator for robustness check, the *Good Faith* indicator still loads negatively, whereas the coefficients on the *Implied Contract* indicator become indistinguishable from zero. Thus, these results are qualitatively similar to those found at the state level: Out of the three WDL exceptions, the good faith exception depresses VC investment, the implied contract does not exhibit a robust effect, and the public policy exception does not impact VC investment.

5.2. Dynamic Effects of WDLs

In this section, we examine the relative timing of WDL adoption and changes in VC investment, again focusing on observations at the state level. Again, we employ a set of timing indicator variables, *Good Faith*^{−1}, *Good Faith*⁰, *Good Faith*¹, and *Good Faith*²⁺, to replace the good faith indicator in the difference-in-differences regressions. We similarly use four indicators, *Implied Contract*^{−1}, *Implied Contract*⁰, *Implied Contract*¹, and *Implied Contract*²⁺ to replace the implied contract indicator which loads negatively at the state-sector analysis. We omit any timing indicators for public policy WDLs because our previous results consistently indicate no effect whatsoever. Table 9 reports the estimation results.

The coefficients on *Good Faith*^{−1} are not distinct from zero, whereas those on *Good Faith*⁰, *Good Faith*¹, and *Good Faith*²⁺ load negatively, regardless of the measure of VC investment and model specification. This shows that at the sector level, VC investment declines as and after, but not before, the good faith exception is adopted. Our previous results suggest that, compared with the GF case, the IC provision involves weaker and/or mixed results. Table 9 finds similar results. The coefficients are generally negative, but in each case smaller than their GF counterpart.

5.3. Which Sectors Are Most Affected by WDLs?

WDLs potentially have differential impacts on different sectors. For example, energy, and transportation are capital intensive. Their low dependency on labor implies that little of their market value depends on their ability to hire or fire employees at will. Other sectors

Table 8. WDLs and VC Investment at the State-Sector Level

Independent variables	Number of VC deals			Dollar Amount of VC Deals		
	OLS	OLS	dC-D'H	OLS	OLS	dC-D'H
<i>Good Faith</i>	−0.076*** (0.021)	−0.061*** (0.023)	−1.033*** (0.378)	−0.177*** (0.036)	−0.146*** (0.039)	−2.019*** (0.683)
<i>Implied Contract</i>	−0.065*** (0.016)	−0.047*** (0.016)	0.029 (0.134)	−0.114*** (0.027)	−0.075*** (0.028)	0.097 (0.197)
<i>Public Policy</i>	−0.007 (0.016)	0.004 (0.017)	0.034 (0.164)	−0.029 (0.028)	−0.001 (0.029)	0.023 (0.260)
Control variables	No	Yes	Yes	No	Yes	Yes
State fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Sector-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	18,000	18,000	18,000	18,000	18,000	18,000
R ²	0.371	0.377		0.356	0.363	

Notes. This table presents the DiD estimates of the effects of good faith, implied contract, and public policy exceptions to at-will employment on state-sector level VC investment in the 1980–2003 period. The dependent variables are $\ln(1 + \text{Number of VC deals})$ and $\ln(1 + \text{Dollar amount of VC deals})$ in a state-sector-year. *Good Faith* is an indicator variable set to one if the state has passed the good faith exception by the year and zero otherwise; *Implied Contract* and *Public Policy* are indicators defined similarly. State characteristics, as defined as in Table 1, are used in some specifications as control variables. State and sector-year fixed effects are controlled for in all specifications. The baseline estimator is OLS, and the de Chaisemartin and D'Haultfoeuille estimator is employed for robustness checks. For the dC-D'H estimates, the coefficients on the three WDL exceptions are estimated separately, controlling for the other two treatments. Standard errors are clustered at the state level and reported in parentheses.

*** marks statistical significance at the 1% level.

Table 9. Dynamic Effects of WDL Adoption on VC Investment at the State-Sector Level

Independent variables	Number of VC deals		Amount of VC deals	
	(1)	(2)	(1)	(2)
<i>Good Faith</i> ⁻¹	-0.041 (0.047)	-0.033 (0.047)	-0.075 (0.081)	-0.068 (0.081)
<i>Good Faith</i> ⁰	-0.182*** (0.044)	-0.147*** (0.044)	-0.324*** (0.075)	-0.259*** (0.076)
<i>Good Faith</i> ¹	-0.157*** (0.044)	-0.120*** (0.045)	-0.285*** (0.076)	-0.223*** (0.077)
<i>Good Faith</i> ²⁺	-0.133*** (0.025)	-0.106*** (0.028)	-0.286*** (0.042)	-0.235*** (0.047)
<i>Implied Contract</i> ⁻¹	-0.018 (0.028)	-0.009 (0.028)	-0.067 (0.048)	-0.051 (0.048)
<i>Implied Contract</i> ⁰	-0.021 (0.028)	-0.014 (0.028)	-0.044 (0.048)	-0.033 (0.048)
<i>Implied Contract</i> ¹	-0.050* (0.029)	-0.039 (0.029)	-0.109** (0.049)	-0.088* (0.049)
<i>Implied Contract</i> ²⁺	-0.083*** (0.021)	-0.061*** (0.021)	-0.143*** (0.036)	-0.101*** (0.037)
<i>Public Policy</i>	0.006 (0.017)	0.014 (0.017)	-0.005 (0.029)	0.014 (0.029)
Control variables	No	Yes	No	Yes
State fixed effects	Yes	Yes	Yes	Yes
Sector-year fixed effects	Yes	Yes	Yes	Yes
Observations	18,000	18,000	18,000	18,000
R ²	0.372	0.377	0.357	0.364

Notes. This table presents the difference-in-differences estimates of the effects of good faith and implied contract exceptions to at-will employment on the dynamics of VC investment at the state-sector level in the 1980–2003 period. The dependent variables are $\ln(1 + \text{Number of VC deals})$ and $\ln(1 + \text{Dollar amount of VC deals})$ in a state-sector-year. *Good Faith*⁻¹ (*Implied Contract*⁻¹) is an indicator variable set to one if a state will adopt the GF (IC) exception in one year and zero otherwise; *Good Faith*⁰ (*Implied Contract*⁰) is an indicator variable set to one if a state adopts the GF (IC) exception in the current year and zero otherwise; *Good Faith*¹ (*Implied Contract*¹) is an indicator variable set to one if a state adopts the GF (IC) exception in the preceding year and zero otherwise; *Good Faith*²⁺ (*Implied Contract*²⁺) is an indicator variable set to one if a state adopts the GF (IC) exception two or more years ago and zero otherwise; *Public Policy* is an indicator that is equal to one if the state has adopted the public policy exception in year t . State characteristics are controlled for in specification (2). State and sector-year fixed effects are controlled for in all specifications. Standard errors are clustered at the state level and reported in parentheses.

*, **, and *** mark statistical significance at the 10%, 5%, and 1% levels, respectively.

like restaurants, groceries, and retail may represent an intermediate case. They do rely on labor, yet can at least partially substitute toward capital when the costs of labor rise (e.g., using electronic menus instead of cashiers). At the other extreme, consider a firm dependent upon a technical staff to conduct R&D or develop software, and which has few tangible assets. Value of such a firm is composed mostly of human capital and is therefore strongly affected by labor law.

To quantify such distinctions, we consider R&D expenditures as a proxy for the importance of specialized human capital which cannot be substituted with physical capital. We supplement this measure with asset tangibility and market-to-book ratio, both of which are related to growth options. High labor dependency sectors should feature high R&D intensity, low asset tangibility, and high market-to-book ratio. R&D intensity is measured as R&D expenditure divided by sales in the previous year; asset tangibility as plant, property, and equipment divided by total assets; and market-to-book ratio as $(\text{total assets} - \text{book value common equity} + \text{market value common equity})/\text{total assets}$. Whenever R&D expenditure is missing, we set R&D intensity to zero. We estimate these ratios for each public firm in the Compustat database in each year during our sample period of 1980–2003, and average them across all firms at the sector level. The average ratios are reported in Table 10.

Noticing the wide variation in these ratios across sectors (for instance, asset tangibility is close to 0.2 for information technology and biotechnology, but around 0.7 for energy and transportation), we rank the sectors by the three ratios separately; for tangibility, the ranking is from low to high, and for the other two ratios, the ranking is from high to low. We then sum up the three ranks for each industry and classify the five sectors with a sum below 20, namely, Biotech, Infotech, Health, Electronics, and Communications, into the “high” labor dependency group, the next five into the “intermediate” labor dependency group, and the last five with a sum above 30 into the “low” labor dependency group.

We then estimate model (1) in each sector and sector group separately and report the coefficients of the *Good Faith* indicator in Table 11. Like in our baseline investigation, the dependent variable is either count or amount of VC deals in a state-year in a sector (or sector group). For a sector group, we aggregate the count (and amount) of VC deals across the sectors. In panel A, the coefficient is negative and statistically significant for the following sectors: Biotech, Communications, Infotech, and Services; the Health sector also obtains negative coefficients with the one in the VC amount regression being statistically significant. These sectors, with the exception of Services, fall in the high labor dependency category. In panel B, the *Good Faith* indicator receives negative and statistically significant coefficients in the high labor dependency group only.

Thus, the good faith exception depresses VC investment only in sectors with high labor dependency. This finding indicates for start-up firms that rely on intangible knowledge assets take the blow as WDLs inflate the labor costs and restrict the operating flexibility, becoming less appealing to both entrepreneurs and investors. Brick-and-mortar start-ups, however, are largely unimpacted.

Table 10. Sector Characteristics Related to Labor Dependency

EVCA sector	R&D		Tangibility		Market-to-book		Sum(rank)	Labor dependency
	Average	Rank	Average	Rank	Average	Rank		
Biotech	0.696	1	0.202	2	2.911	1	4	High
Infotech	0.168	2	0.176	1	2.428	2	5	High
Health	0.137	3	0.280	5	2.255	3	11	High
Electronics	0.106	4	0.249	3	1.733	6	13	High
Communications	0.061	5	0.337	8	1.840	4	17	High
Agriculture	0.045	6	0.503	12	1.779	5	23	Intermediate
Services	0.004	12	0.253	4	1.628	8	24	Intermediate
Industrials	0.028	8	0.310	6	1.426	11	25	Intermediate
Consumers	0.007	10	0.379	9	1.696	7	26	Intermediate
Chemicals	0.033	7	0.410	11	1.552	9	27	Intermediate
BusSupplies	0.015	9	0.404	10	1.378	13	32	Low
Construction	0.005	11	0.330	7	1.328	14	32	Low
Energy	0.003	13	0.776	15	1.517	10	38	Low
Transportation	0.001	14	0.675	13	1.410	12	39	Low
Other	0.001	15	0.762	14	1.294	15	44	Low

Notes. This table presents financial variables of public firms in EVCA sectors that are related to labor dependency, including R&D intensity, asset tangibility, and market-to-book ratio. The sample is all the publicly listed firm in the Compustat database during the 1980–2003 period. R&D intensity is measured as R&D expenditure divided by previous-year sales. Asset tangibility is property, plant, and equipment as a ratio of the beginning-of-year assets. Market-to-book ratio is estimated as (assets – common equity + market capitalization of shares)/assets. The three variables are considered negatively, positively, and positively related to labor dependency, respectively. Average R&D, tangibility, and market-to-book in each EVCA sector and the sector rank in the respective characteristic are reported. The sum of the three ranks is then used to classify the sectors into high, intermediate, and low labor dependency groups.

Table 11. Effect of Good Faith Exception in Separate Sectors

Panel A. Effect in sectors		
Sector	<i>N</i> (VC deals)	<i>\$</i> (VC deals)
Biotech	−0.272***	−0.548***
Infotech	−0.225***	−0.416***
Health	−0.094	−0.271**
Electronics	−0.023	−0.049
Communications	−0.245***	−0.591***
Agriculture	−0.032	−0.028
Services	−0.178**	−0.406***
Industrials	0.048	0.114
Consumers	−0.009	−0.066
Chemicals	0.016	0.029
BusSupplies	−0.053	−0.161*
Construction	0.016	0.068
Energy	0.098***	0.159**
Transportation	−0.002	−0.050
Other	0.040	0.020
Panel B. Effect in sector groups		
Labor dependency	<i>N</i> (VC deals)	<i>\$</i> (VC deals)
High	−0.324***	−0.612***
Intermediate	−0.096	−0.177
Low	0.076	0.061

Notes. This table presents the DiD estimates of the effect of good faith exceptions on state-level VC investment in each EVCA sector (panel A) and sector groups based on labor dependency (panel B). In panel A, sectors are listed in the descending order of labor dependency based on R&D intensity, asset tangibility, and market-to-book ratio of public firms. In panel B, sectors are classified into the high, intermediate, and low labor dependency groups, as reported in Table 10. The dependent variables are $\ln(1 + \text{Number of VC deals})$ and $\ln(1 + \text{Dollar amount of VC deals})$ in a state-sector-year. *Good Faith* is an indicator variable set to one if the state has passed the good faith exception by the year and zero otherwise. State and sector-year fixed effects are controlled for. Standard errors are clustered at the state level.

*, **, and *** mark statistical significance at the 10%, 5%, and 1% levels, respectively.

6. Conclusion

This paper studies the effect of WDLs which impair firms' operating flexibility by making it costlier to fire employees. Start-ups in particular are affected because they depend upon operating flexibility to take advantage of growth options.

Our evidence shows that the passage of the WDLs causes a marked decline in venture capital activity. We document this effect using a difference-in-differences framework made possible by the staggered adoption of the WDLs between 1970 and 1998. Our conclusions are supported by several additional tests: (i) examination of the dynamic effects of WDLs indicates VC investment declines after the adoption but not before, (ii) survival analysis based on the Cox proportional hazard model is inconsistent with reverse causality, and (iii) we generate counterfactual samples by randomly assigning WDL exceptions to states in random years.

The effect primarily comes from the good faith exemption, which is considered by legal scholars to be the strongest type of WDL. Its negative effect concentrates in a few sectors with high labor dependency such as biotechnology, information technology and communications. The implied contract exception exhibits a negative effect in some of our investigations, but the effect

is largely insignificant. The public policy exemption, considered the weakest form of WDL, has no effect on venture capital activity.

In addition, VC investment declines despite the increase in firm creation. We replicate the Acharya et al. (2014) results that the good faith exception is associated with an increase in number of new establishments and firms, and further show that out of these new businesses, the fraction that received VC funding substantially declined. Furthermore, the heightened business creation is observed only in the smallest size buckets and non-high-tech industries. This indicates that WDLs encourage the creation of conventional firms and small independent shops (including self-employment) but not firms suitable for VC investment, that is, firms that heavily depend upon growth options and intellectual capital. The distinction employed in this paper—VC financed versus non-VC financed—is a rather coarse one, and future work may provide a more complete description of which firm types are hindered by WDLs. In addition, our work does not quantify the extent to which the dampening in VC-backed firm creation comes from the supply side (willingness of investors to supply funds) versus the demand side (willingness of entrepreneurs to create firms).

Appendix. Variables

Variable	Definition
Venture capital investment	
<i>Number of VC deals</i>	Total number of VC deals in a given state and year. In regressions, take the logarithm of one plus the number of VC deals.
<i>Amount of VC investment</i>	Total dollar amount of VC deals in a given state and year, measured in million U.S. dollars, adjusted by the Consumer Price Index (= 100 in year 2000). In regressions, take the logarithm of one plus the amount.
WDL measures	
<i>Good Faith (GF)</i>	A dummy variable equal to one if a state has adopted the good faith exception in a given year and zero otherwise
<i>Implied Contract (IC)</i>	A dummy variable equal to one if a state has adopted the implied contract exception in a given year and zero otherwise
<i>Public Policy (PP)</i>	A dummy variable equal to one if a state has adopted the public policy exception in a given year and zero otherwise
State characteristics	
<i>Population</i>	Population in a given state and year, in millions. The logarithm of the value is used in regressions.
<i>GDP</i>	Gross domestic product of a state in a given year, in constant billion dollars of year 2000. The logarithm of the value is used in regressions.
<i>Real GDP growth</i>	Percentage growth rate of state GDP in a given year
<i>Unemployment insurance</i>	Maximum total potential benefit available under the unemployment insurance system in a given state and year. The logarithm is used in regressions.
<i>Homestead exemptions</i>	The amount of homestead exemptions in a state; adjust the original data by inflation and then apply the $\log(1 + x)$ transformation
<i>Political balance</i>	Ratio of the Democrat-to-Republican representatives in the state House of Representatives for a given state and year
<i>Religiosity</i>	Number of adherents to recognized religious denominations in each state every 1,000 people. Original data are from the Association of Religion Data Archives' decennial surveys in 1980, 1990, 2000, and 2010. We obtain individual years' missing data through linear interpolation and apply log transformation to the data.
<i>FOIA</i>	A dummy variable that is equal to one after a state adopts the Freedom of Information Act and zero otherwise
<i>Noncompete index</i>	An index of enforceability of covenants not to compete following Garmaise (2012) and Bird and Knopf (2015)

Appendix. (Continued)

Variable	Definition
<i>SBIR</i>	The amount of Small Business Innovation Research awards in a state in a given year; adjust the original data by inflation and then apply the log transformation
<i>College enrollment</i>	Number of students enrolled in degree-granting higher education institutions in a given state five years ago, in millions
More variables in hazard model	
<i>Circuit Good Faith</i>	The fraction of other states in the same federal circuit region that have adopted the good faith exception in a given year
<i>Circuit Implied Contract</i>	The fraction of other states in the same federal circuit region that have adopted the implied contract exception in a given year
<i>Circuit Public Policy</i>	The fraction of other states in the same federal circuit region that have adopted the public policy exception in a given year
<i>Union membership density</i>	The fraction of nonagricultural employees in a state who are union members in a given year
<i>Unemployment rate</i>	The fraction of workers in a state who are in the labor force but unemployed in a given year
Δ <i>Union membership density</i>	Change in union membership density in a given state from the previous year
Δ <i>Unemployment rate</i>	Change in unemployment rate in a state from the previous year

Endnotes

- ¹ In what follows, we interchangeably describe the effect of WDLs as (i) increasing firing costs, (ii) increasing labor costs, (iii) reducing labor flexibility, or (iv) reducing operating flexibility. Point (ii) follows from (i) because the present value of obligations to a given employee include the expected future costs of dismissal.
- ² The law and finance literatures (e.g., La Porta et al. 1998) categorize four legal families in Europe: English, French, German, and Scandinavian. The English law, also known as common law, is law developed by judges through decisions of courts; the French, German, and Scandinavian laws all fall in the category of civil law, which features codified statutes. The U.S. law is common law.
- ³ The identification of passage of WDLs follows Serfling (2016, p. 2247). New Hampshire and Oklahoma enacted the good faith exception but then reversed it several years later; Arizona and Missouri reversed the implied contract exception in the 1980s.
- ⁴ “Prior to 1990, insurers offered EPLI on a limited basis. With the increase in claims involving wrongful employment practices, the number of insurers offering an EPLI product ballooned” (Davis and Gassman 2013, p. 23).
- ⁵ We thank Professor Bird for generously providing the noncompete enforceability index data.
- ⁶ We thank Professor Hynes (Hynes et al. 2004) for kindly providing the homestead exemption data.
- ⁷ For detailed description of the data, please see Gao et al. (2017). For the variable construction please see Chircop et al. (2020).
- ⁸ Following Hynes et al. (2004), unlimited homestead exemptions are replaced with the average of the highest monetary limits for homestead exemptions among other states in the same year. Yet, it does not carry merit to report the mean, standard deviation, or maximum of the variable.
- ⁹ When we cluster the standard errors at the state and year level, we obtain similar results.
- ¹⁰ Percentage change is computed as $[exp(\beta_1) - 1] \times 100$. Some states do not have VC investment in certain years; hence, our logarithm transformation is $\ln(1 + VC \text{ count})$ instead of $\ln(VC \text{ count})$, which makes our reported percent change an approximate estimate. A similar caveat applies to the reported percentage change in amount of VC investment.
- ¹¹ For a summary of the literature on the TWFE research design issues and proposed estimators, please see de Chaisemartin and D’Haultfoeuille (2023) and Roth et al. (2023).

- ¹² We use the *did_multiplegt* command by de Chaisemartin and D’Haultfoeuille (2020) in Stata to estimate the effects of each WDL exception in the maximum possible number of posttreatment periods and report the *average_effect* that represents the average effect of staying treated.
- ¹³ When we replace either the *Implied Contract* or *Public Policy* indicator with four timing indicators similarly defined, none of the indicators loads in any specification.
- ¹⁴ We thank a referee for this insightful observation.
- ¹⁵ For our purposes, this list contains both type 1 and type 2 errors. For example, the list does not separately delineate VC-focused industries such as biotechnology. On the other hand, a self-employed person doing web design consulting would count as “high tech” yet would be unsuitable for VC investment.
- ¹⁶ For an EVCA sector definition, please see Bozkaya and Kerr (2014). We combine Industrial Products and Services and Industrial Automation sectors into the Industrials sector to minimize potential classification errors. Our sample does not include financial services companies. For brevity, we assign shorter names to these sectors, for example, Health to Medical/Health Related, Infotech to Computer Related, Electronics to Other Electronics Related, and Business Supplies to Other Manufacturing that covers mainly business products and supplies. To our knowledge there exists no similar sector classification specifically for venture capital investment in the United States. Yet when we use the Fama–French 17 industry classification to define sectors, results are qualitatively similar.

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